INTERNATIONAL MATHEMATICAL UNION

FIELDS MEDAL 2026

HONG WANG

SHORT CITATION

For her work in harmonic analysis and geometric measure theory, including applications of multiscale and decoupling techniques to the local smoothing conjecture for the planar wave equation, and major advances in Fourier restriction, Falconer distance sets, Furstenberg sets in the plane, and the Kakeya problem in three dimensions.

LONG CITATION

Extending and refining modern techniques in harmonic analysis and geometric measure theory, Hong Wang has achieved a series of major breakthroughs on some of the most central problems in analysis, combining multiscale methods with deep geometric and combinatorial insights.

Wang has played a decisive role in advancing the theory of local smoothing for wave equations. Together with Larry Guth and Ruixiang Zhang, she introduced powerful decoupling and multiscale arguments to resolve the local smoothing conjecture for the planar wave equation.

Her work has also led to fundamental progress in Fourier restriction theory and in geometric problems related to distance sets. In particular, with Guth, Alex Iosevich, and Yumeng Ou, she obtained major results on the Falconer distance set problem, clarifying the relationship between Hausdorff dimension and the structure of distance distributions. In addition, with Kevin Ren, she made breakthrough contributions to the theory of Furstenberg sets in the plane, significantly deepening the understanding of directional phenomena in fractal geometry.

In higher dimensions, Wang has achieved major advances on the Kakeya problem in three dimensions. In joint work with Joshua Zahl, she developed new bounds and structural techniques for collections of thin tubes, opening new directions and reshaping current approaches to this central problem. More broadly, her work has introduced methods that are now central to contemporary harmonic analysis and continue to influence a wide range of problems in analysis, geometry, and beyond.